Nous avons utilisé la commande `tcpdump -i eth0` pour capturer les paquets réseau.

```
ubuntu@RiadhBouhamed:~$ sudo tcpdump -i eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:34:46.271064 IP 172.23.172.28.51023 > prod-ntp-4.ntp1.ps5.canonical.com.n
tp: NTPv4, Client, length 48
14:34:46.331110 IP prod-ntp-4.ntp1.ps5.canonical.com.ntp > 172.23.172.28.510
23: NTPv4, Server, length 48
14:34:52.028872 ARP, Request who-has RiadhBouhamed.mshome.net tell 172.23.17
2.28, length 28
14:34:52.029867 ARP, Reply RiadhBouhamed.mshome.net is-at 00:15:5d:c9:08:94
(oui Unknown), length 28
14:35:19.028617 IP 172.23.172.28.56056 > prod-ntp-4.ntp1.ps5.canonical.com.n
tp: NTPv4, Client, length 48
14:35:19.082449 IP prod-ntp-4.ntp1.ps5.canonical.com.ntp > 172.23.172.28.560
56: NTPv4, Server, length 48
14:35:24.404966 ARP, Request who-has 172.23.172.28 (00:15:5d:be:ef:c9 (oui U
nknown)) tell RiadhBouhamed.mshome.net, length 28
14:35:24.404988 ARP, Reply 172.23.172.28 is-at 00:15:5d:be:ef:c9 (oui Unknow
n), length 28
14:35:24.862293 ARP, Request who-has RiadhBouhamed.mshome.net tell 172.23.17
2.28, length 28
14:35:24.863042 ARP, Reply RiadhBouhamed.mshome.net is-at 00:15:5d:c9:08:94
(oui Unknown), length 28
14:35:31.518040 IP6 fe80::215:5dff:febe:efc9 > ip6-allrouters: ICMP6, router
 solicitation, length 16
14:35:51.856744 IP 172.23.172.28.42230 > prod-ntp-4.ntp1.ps5.canonical.com.n
tp: NTPv4, Client, length 48
14:35:51.924852 IP prod-ntp-4.ntp1.ps5.canonical.com.ntp > 172.23.172.28.422
30: NTPv4, Server, length 48
14:35:57.430748 ARP, Request who-has RiadhBouhamed.mshome.net tell 172.23.17
2.28, length 28
14:35:57.431115 ARP, Reply RiadhBouhamed.mshome.net is-at 00:15:5d:c9:08:94
(oui Unknown), length 28
^C
15 packets captured
15 packets received by filter
0 packets dropped by kernel
```

Nous avons observé plusieurs types de paquets :

- **NTP** : échanges entre la machine locale et un serveur de temps pour synchronisation
- **ARP** : résolution d'adresses IP en adresses MAC
- **ICMPv6** : messages de découverte de routeur

Ces observations montrent que le réseau fonctionne en échangeant différents types de paquets selon les besoins (résolution d'adresse, synchronisation, routage, etc.).

✍️ Analyse TCP/HTTP

Nous avons utilisé `tcpdump` pour analyser une communication HTTP avec un serveur distant.

Commandes utilisées

```
sudo tcpdump -i eth0 host google.com and port 80
```

Ce que nous écrivons :

- démarrer la capture filtrée sur HTTP (port 80)

Ce que nous observons :

- les paquets TCP entre notre machine et le serveur Google

  Après établissement de la connexion TCP, nous avons observé l'envoi de la requête HTTP (GET).

Commandes utilisées

```
telnet google.com 80
```

Puis dans telnet :

```
GET / HTTP/1.1
Host: google.com
```

Ce que nous écrivons :

- une requête HTTP manuelle

Ce que nous observons dans tcpdump :

- des paquets avec flags `[S]`, `[S.]`, puis `[P.]`

  Les données ont été transmises sous forme de plusieurs segments TCP (flags PSH, ACK), ce qui montre que TCP découpe les données en paquets.

Ce que nous observons :

```
Flags [P.]
length 23
length 16
length 18
```

Chaque segment envoyé est suivi d'un accusé de réception (ACK) du serveur, garantissant la fiabilité de la transmission.

Ce que nous observons :

Flags [.]

Enfin, la connexion est fermée à l'aide des segments FIN/ACK, montrant la terminaison correcte de la communication TCP.

Ce que nous observons :

Flags [F.]

Flags [.]

💡 Bonus

L'utilisation de telnet a entraîné l'envoi fragmenté de la requête HTTP, ce qui a permis d'observer clairement le mécanisme de segmentation TCP.

```
ubuntu@RiadhBouhamed:~$ telnet google.com 80
Trying 142.251.39.174...
Connected to google.com.
Escape character is '^]'.
GET / HTTP/1.1
Host: google.com
 telnet google.com 80
GET / HTTP/1.1
Host: google.com
^C^CConnection closed by foreign host.
```

```
ubuntu@RiadhBouhamed:~$ sudo tcpdump -i eth0 host google.com and port 80
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:47:41.900850 IP 172.23.172.28.51252 > ncmrsb-ak-in-f14.1e100.net.http: Fl
ags [P.], seq 990603124:990603147, ack 1082584330, win 504, options [nop,nop
,TS val 2627156276 ecr 3050357305], length 23: HTTP
14:47:41.933115 IP ncmrsb-ak-in-f14.1e100.net.http > 172.23.172.28.51252: Fl
ags [.], ack 23, win 1050, options [nop,nop,TS val 3050417223 ecr 2627156276
], length 0
14:47:52.940747 IP 172.23.172.28.51252 > ncmrsb-ak-in-f14.1e100.net.http: Fl
ags [P.], seq 23:39, ack 1, win 504, options [nop,nop,TS val 2627167316 ecr
3050417223], length 16: HTTP: GET / HTTP/1.1
14:47:52.983261 IP ncmrsb-ak-in-f14.1e100.net.http > 172.23.172.28.51252: Fl
ags [.], ack 39, win 1050, options [nop,nop,TS val 3050428498 ecr 2627167316
], length 0
14:47:59.879508 IP 172.23.172.28.51252 > ncmrsb-ak-in-f14.1e100.net.http: Fl
ags [P.], seq 39:57, ack 1, win 504, options [nop,nop,TS val 2627173435 ecr
3050428498], length 18: HTTP
14:47:59.925074 IP ncmrsb-ak-in-f14.1e100.net.http > 172.23.172.28.51252: Fl
ags [.], ack 57, win 1050, options [nop,nop,TS val 3050434725 ecr 2627173435
], length 0
14:48:14.007725 IP 172.23.172.28.51252 > ncmrsb-ak-in-f14.1e100.net.http: Fl
ags [P.], seq 57:62, ack 1, win 504, options [nop,nop,TS val 2627187564 ecr
3050434725], length 5: HTTP
14:48:14.060452 IP ncmrsb-ak-in-f14.1e100.net.http > 172.23.172.28.51252: Fl
ags [.], ack 62, win 1050, options [nop,nop,TS val 3050449302 ecr 2627187564
], length 0
14:48:18.464321 IP 172.23.172.28.51252 > ncmrsb-ak-in-f14.1e100.net.http: Fl
ags [P.], seq 62:67, ack 1, win 504, options [nop,nop,TS val 2627192020 ecr
3050449302], length 5: HTTP
14:48:18.508631 IP ncmrsb-ak-in-f14.1e100.net.http > 172.23.172.28.51252: Fl
ags [.], ack 67, win 1050, options [nop,nop,TS val 3050453873 ecr 2627192020
], length 0
14:48:27.759235 IP ncmrsb-ak-in-f14.1e100.net.http > 172.23.172.28.51252: Fl
ags [F.], seq 1, ack 67, win 1050, options [nop,nop,TS val 3050463321 ecr 26
27192020], length 0
14:48:27.759629 IP 172.23.172.28.51252 > ncmrsb-ak-in-f14.1e100.net.http: Fl
ags [F.], seq 67, ack 2, win 504, options [nop,nop,TS val 2627201315 ecr 305
0463321], length 0
14:48:27.791571 IP ncmrsb-ak-in-f14.1e100.net.http > 172.23.172.28.51252: Fl
ags [.], ack 68, win 1050, options [nop,nop,TS val 3050463362 ecr 2627201315
```

```c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

int main() {
    int sock;
    struct sockaddr_in server;
    char buffer[4096];

    char request[] =
        "GET / HTTP/1.1\r\n"
        "Host: example.com\r\n"
        "Connection: close\r\n\r\n";

    sock = socket(AF_INET, SOCK_STREAM, 0);
    if (sock < 0) {
        perror("socket");
        return 1;
    }

    server.sin_family = AF_INET;
    server.sin_port = htons(80);

    if (inet_pton(AF_INET, "104.21.48.1", &server.sin_addr) <= 0) {
        perror("inet_pton");
        return 1;
    }
```

```c
if (connect(sock, (struct sockaddr*)&server, sizeof(server)) < 0) {
    perror("connect");
    return 1;
}

printf("Connected\n");

if (send(sock, request, strlen(request), 0) < 0) {
    perror("send");
    return 1;
}

printf("Request sent\n");

int n;
while ((n = recv(sock, buffer, sizeof(buffer)-1, 0)) > 0) {
    buffer[n] = '\0';
    printf("%s", buffer);
}

printf("\nDone\n");

close(sock);
return 0;
```

```
ubuntu@RiadhBouhamed:/mnt/c/Users/riadb/OneDrive/Desktop/tp reseau$ gcc http_client.c -o http_client
ubuntu@RiadhBouhamed:/mnt/c/Users/riadb/OneDrive/Desktop/tp reseau$ ./http_client
Connected
Request sent
HTTP/1.1 200 OK
Date: Mon, 13 Apr 2026 14:26:11 GMT
Content-Type: text/html
Transfer-Encoding: chunked
Connection: close
Server: cloudflare
Age: 1678
cf-cache-status: HIT
Age: 1678
cf-cache-status: HIT
CF-RAY: 9ebb24f7beb3e219-MRS

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<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width,
 initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-s
ize:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domai
n is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://i
ana.org/domains/example">Learn more</a></p></div></body></html>

0


Done
```